

동역학

(Engineering Mechanics: Dynamics)

Chapter 12-1

영남대학교 기계공학부 김병일 교수

수업계획서

교수명

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강의 시간

(0834) **월요일 10:30 - 11:45, 수요일 09:00 - 10:15** (ME Building 160)

면담 시간

Anytime available

강의 소개

- 역학의 원리를 이해하고
- 운동과 힘의 관계를 수학적으로 표현하며
- 수식의 결과를 이해하여
- 기계 시스템을 동역학적으로 파악하는 능력을 배양한다

강의 목표

- 힘을 받는 물체의 운동을 이해하고 기계 시스템에서 발생하는 운동학 및 운동 역학적 문제를 해석할 수 있는 능력을 배양한다.
- 질점과 강체의 운동, 운동역학을 공부하면서 동적 평형, 일과 에너지, 충격량 및 운동량 등의 원리를 학습한다.



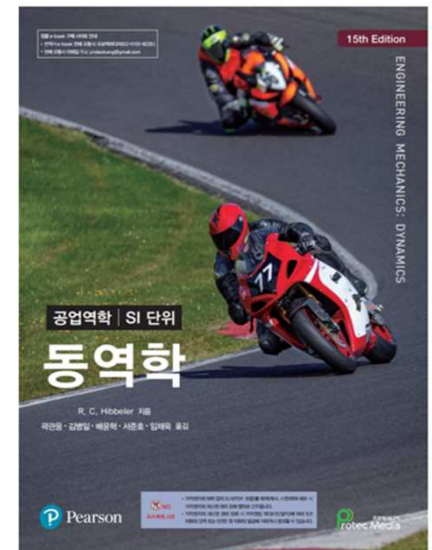
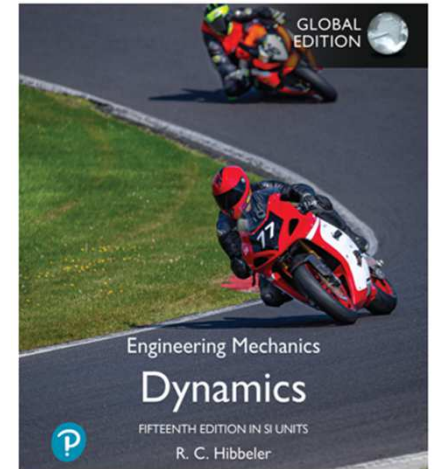
수업계획서 (계속)

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학습평가

중간고사	40%
기말고사	40%
과제물	15%
출석	5%

과제물 모든 과제물 보고서는 정해진 시간까지 제출해야 함.
 늦게 낼 시 감점.



Course Contents

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3	Chapter 12. Kinematics of Particles	질점의 동역학	
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Engineering Mechanics: Dynamics

Fifteenth Edition in SI Units



CHAPTER 12

KINEMATICS
OF A
PARTICLE

What is Dynamics?

Definition

- Kinematic relationships are used to help us determine the trajectory of a snowboarder completing a jump, the orbital speed of a satellite, and accelerations during acrobatic flying.

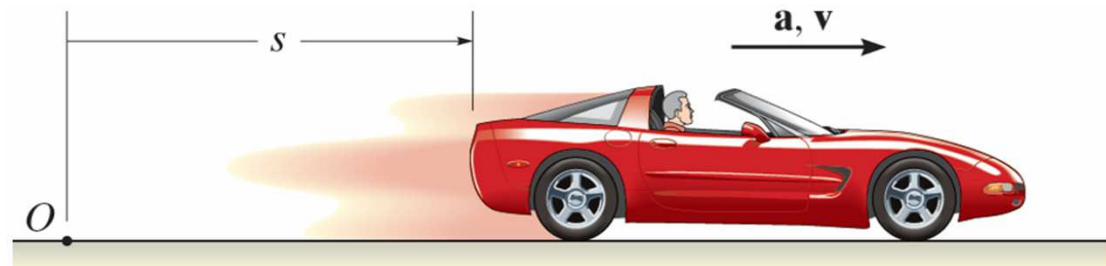


Applications

- The motion of large objects, such as rockets, airplanes, or cars, can often be analyzed as if they were particles.
- Why?
- If we measure the altitude of this rocket as a function of time, how can we determine its velocity and acceleration?



- A sports car travels along a straight road.
- Can we treat the car as a particle?
- If the car accelerates at a constant rate, how can we determine its position and velocity at some instant?

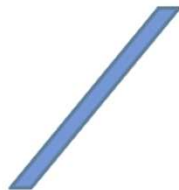


Particle vs. Rigid Body

- **Particles** are bodies where all the mass is concentrated at a single point in space. Particle analysis will only have to take into account the forces acting on the body and translational motion because rotation is not considered for particles.

particle •

Rigid-body (continuum of particles)



- **Rigid bodies** on the other hand have mass that is distributed throughout a finite volume. Rigid body analysis is more complex and also has to take into account moments and rotational motions. In actuality, no bodies are truly particles, but some bodies can be approximated as particles to simplify analysis. Bodies are often assumed to be particles if the rotational motions are negligible when compared to the translational motions, or in systems where there is no moment exerted on the body such as a concurrent force system.



An Overview of Mechanics

- **Kinematics:** study of the geometry of motion. Relates displacement, velocity, acceleration, and time *without reference* to the cause of motion.
- **Kinetics:** study of the relations existing between the forces acting on a body, the mass of the body, and the motion of the body. Kinetics is used to predict the motion caused by given forces or to determine the forces required to produce a given motion.

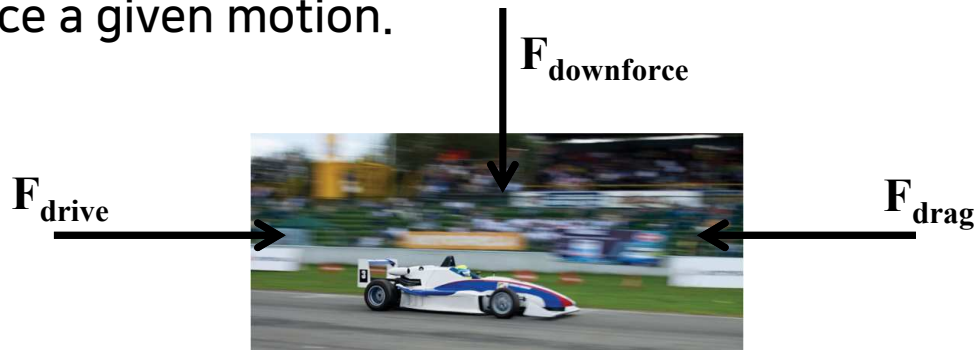
Mechanics: The study of how bodies react to the forces acting on them.

Statics: The study of bodies in equilibrium.

Dynamics:

1. **Kinematics** - concerned with the geometric aspects of motion

2. **Kinetics** - concerned with the forces causing the motion



Particle Kinematics

- **Rectilinear motion**: position, velocity, and acceleration of a particle as it moves along a straight line.

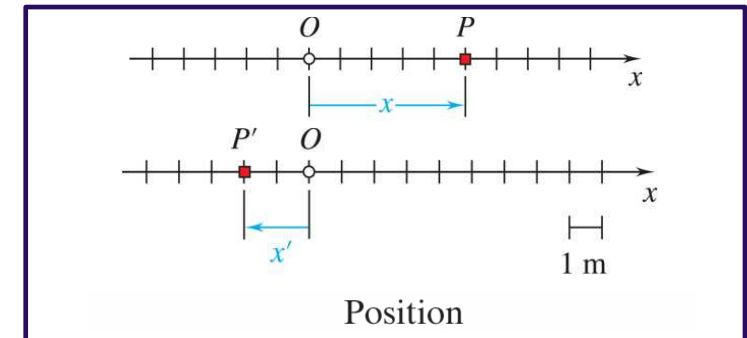


- **Curvilinear motion**: position, velocity, and acceleration of a particle as it moves along a curved line in two or three dimensions.



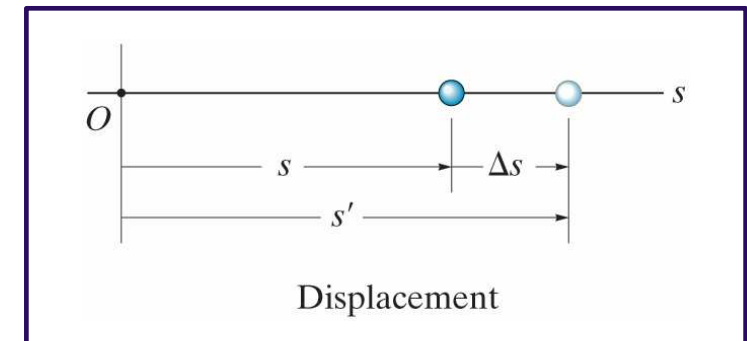
Rectilinear Kinematics: Continuous Motion

- A particle travels along a straight-line path defined by the coordinate axis s . **(Position Coordinate)**
- The **position** of the particle at any instant, relative to the origin, O , is defined by the position vector r , or the scalar s . Scalar s can be positive or negative.



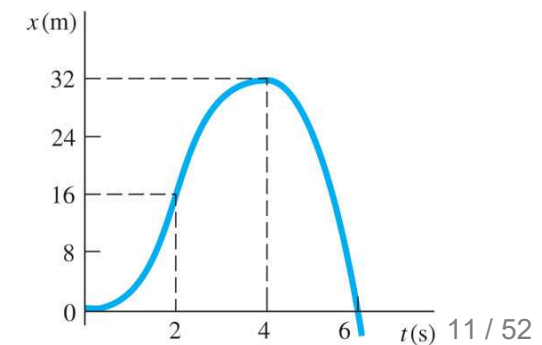
- The **displacement** is defined as the change in position.

Vector form: $\Delta r = r' - r$ Scalar form: $\Delta s = s' - s$



- The **motion of a particle is known** if the position coordinate is known for every value of time t .
- May be expressed in the form of a function, e.g., or in the form of a graph x vs. t .

$$x = 6t^2 - t^3$$



Velocity

- **Velocity** is a measure of the rate of change in the position of a particle. It is a vector quantity (it has both magnitude and direction). The magnitude of the velocity is called speed, with units of m/s.

- The **average velocity** of a particle during a time interval Δt is

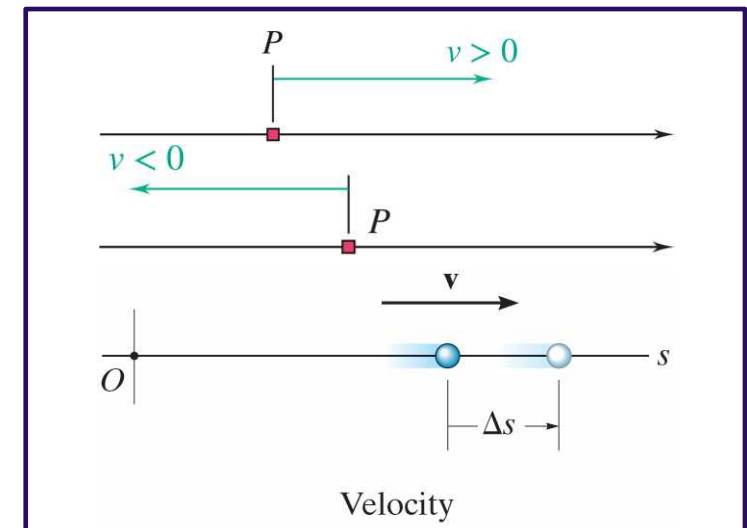
$$v_{avg} = \Delta r / \Delta t$$

- The **instantaneous velocity** is the time-derivative of position.

$$v = dr/dt$$

- **Speed** is the magnitude of velocity:

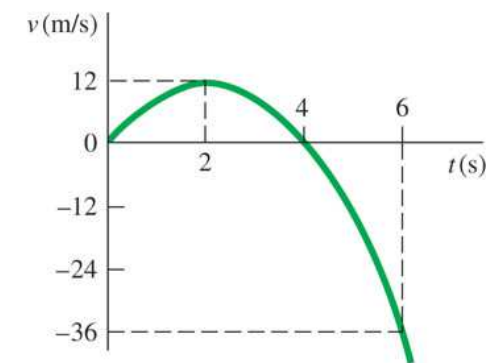
$$v = ds/dt$$



$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

$$x = 6t^2 - t^3$$

$$v = \frac{dx}{dt} = 12t - 3t^2$$



Acceleration

- **Acceleration** is the rate of change in the velocity. It is a vector quantity. Typical units are m/s^2
- The **instantaneous acceleration** is the time derivative of velocity.

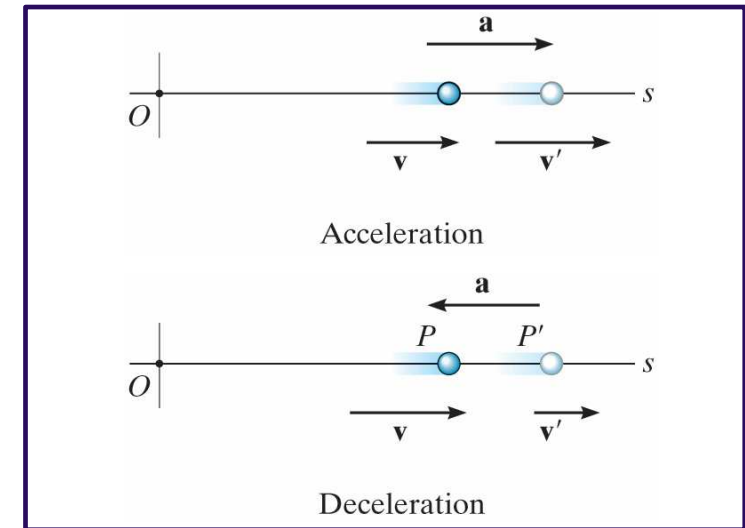
Vector form: $a = dv/dt$

Scalar form: $a = dv / dt = d^2s / dt^2$

positive: increasing positive velocity or decreasing negative velocity
negative: decreasing positive velocity or increasing negative velocity.

- Acceleration can be positive (speed increasing) or negative (speed decreasing).
- The derivative equations for velocity and acceleration can be manipulated to get

$$a ds = v dv$$

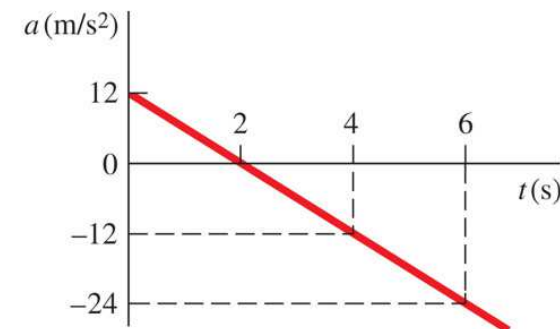


$$a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t}$$

$$= \frac{dv}{dt} = \frac{d^2x}{dt^2}$$

$$v = 12t - 3t^2$$

$$a = \frac{dv}{dt} = 12 - 6t$$



Summary of Kinematic Relations: Rectilinear Motion

- Differentiate position to get velocity and acceleration.

$$v = ds/dt; \quad a = dv/dt \text{ or } a = v dv/ds$$

- Integrate acceleration for velocity and position.

$$\text{Velocity: } \int_{v_0}^v dv = \int_0^t a dt \text{ or } \int_{v_0}^v v dv = \int_{s_0}^s ds \quad \text{Position: } \int_{s_0}^s ds = \int_0^t v dt$$

- Note that s_0 and v_0 represent the initial position and velocity of the particle at $t = 0$.



Constant Acceleration

The three kinematic equations can be integrated for the special case when **acceleration is constant ($a = a_c$)** to obtain very useful equations. A common example of constant acceleration is gravity; i.e., a body freely falling toward earth. In this case, $a_c = g = 9.81 \text{ m/s}^2 = 32.2 \text{ ft/s}^2$ downward. These equations are:

$$\int_{v_0}^v dv = \int_0^t a_c dt \quad \text{yields} \quad v = v_0 + a_c t$$

$$\int_{s_0}^s ds = \int_0^t v dt \quad \text{yields} \quad s = s_0 + v_0 t + (1/2) a t^2$$

$$\int v dv = \int a ds \quad \text{yields} \quad v^2 = (v_0)^2 + 2a(s - s_0)$$



Determining the Motion of a Particle

- We often determine accelerations from the forces applied (kinetics will be covered later)
- Generally have three classes of motion
 1. acceleration given as a function of time, $a = f(t)$
 2. acceleration given as a function of position, $a = f(x)$
 3. acceleration given as a function of velocity, $a = f(v)$

☐ Can you think of a physical example of when force is a function of position? When force is a function of velocity?

A Spring

Drag

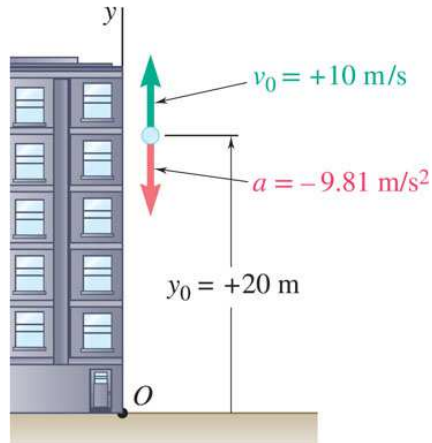


Acceleration as a function of time, position, or velocity

If...	Kinematic relationship	Integrate
$a = a(t)$	$\frac{dv}{dt} = a(t)$	$\int_{v_0}^v dv = \int_0^t a(t)dt$
$a = a(x)$	$dt = \frac{dx}{v}$ and $a = \frac{dv}{dt}$ $\downarrow$ $v dv = a(x)dx$	$\int_{v_0}^v v dv = \int_{x_0}^x a(x)dx$
$a = a(v)$	$\frac{dv}{dt} = a(v)$	$\int_{v_0}^v \frac{dv}{a(v)} = \int_0^t dt$
	$v \frac{dv}{dx} = a(v)$	$\int_{x_0}^x dx = \int_{v_0}^v \frac{v dv}{a(v)}$



Sample Problem: Ball Toss



Ball tossed with 10 m/s vertical velocity from window 20 m above ground.

Determine:

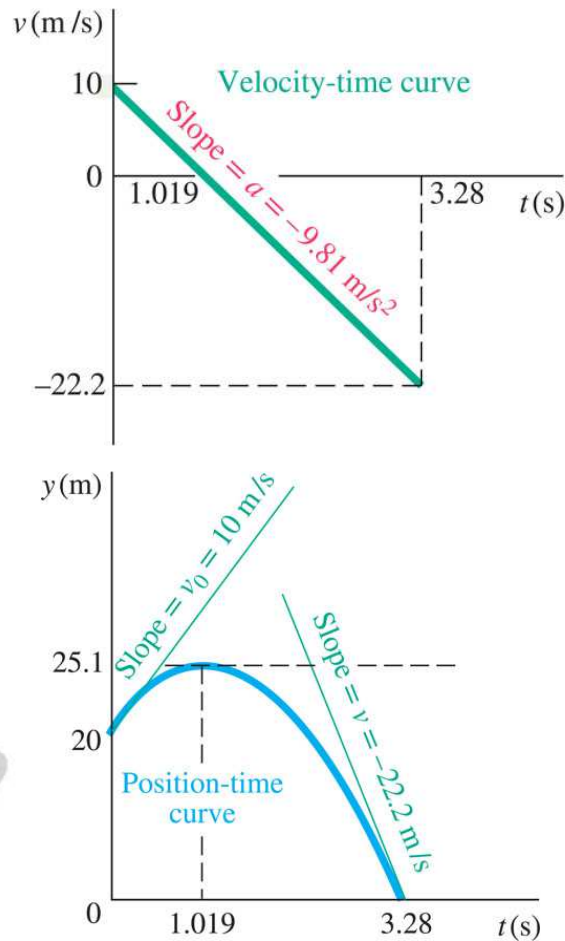
- velocity and elevation above ground at time t ,
- highest elevation reached by ball and corresponding time, and
- time when ball will hit the ground and corresponding velocity.

STRATEGY:

1. Integrate twice to find $v(t)$ and $y(t)$.
2. Solve for t when velocity equals zero (time for maximum elevation) and evaluate corresponding altitude.
3. Solve for t when altitude equals zero (time for ground impact) and evaluate corresponding velocity.



Sample Problem: Ball Toss



MODELING and ANALYSIS:

Integrate twice to find $v(t)$ and $y(t)$.

$$\frac{dv}{dt} = a = -9.81 \text{ m/s}^2$$

$$\int_{v_0}^{v(t)} dv = - \int_0^t 9.81 dt \quad v(t) - v_0 = -9.81t$$

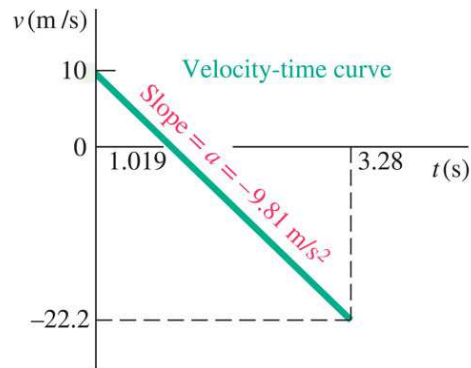
$$v(t) = 10 \frac{\text{m}}{\text{s}} - \left(9.81 \frac{\text{m}}{\text{s}^2} \right) t$$

$$\frac{dy}{dt} = v = 10 - 9.81t$$

$$\int_{y_0}^{y(t)} dy = \int_0^t (10 - 9.81t) dt \quad y(t) - y_0 = 10t - \frac{1}{2} 9.81t^2$$

$$y(t) = 20\text{m} + \left(10 \frac{\text{m}}{\text{s}} \right) t - \left(4.905 \frac{\text{m}}{\text{s}^2} \right) t^2$$

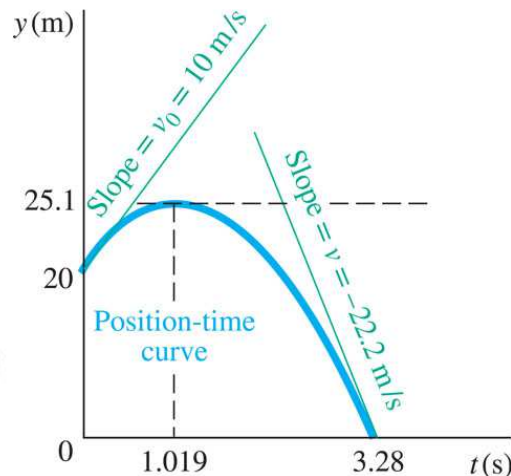
Sample Problem: Ball Toss



- Solve for t when velocity equals zero and evaluate corresponding altitude.

$$v(t) = 10 \frac{\text{m}}{\text{s}} - \left(9.81 \frac{\text{m}}{\text{s}^2} \right) t = 0$$

$$t = 1.019\text{s}$$



- Solve for t when altitude equals zero and evaluate corresponding velocity.

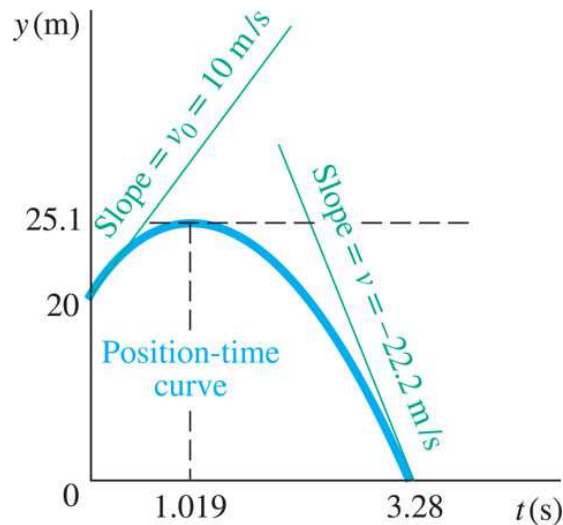
$$y(t) = 20\text{m} + \left(10 \frac{\text{m}}{\text{s}} \right) t - \left(4.905 \frac{\text{m}}{\text{s}^2} \right) t^2$$

$$y = 20\text{m} + \left(10 \frac{\text{m}}{\text{s}} \right) (1.019\text{s})$$

$$y = 25.1\text{m}$$



Sample Problem: Ball Toss



- Solve for t when altitude equals zero and evaluate corresponding velocity.

$$y(t) = 20\text{m} + \left(10 \frac{\text{m}}{\text{s}}\right)t - \left(4.905 \frac{\text{m}}{\text{s}^2}\right)t^2 = 0$$

$$t = -1.243\text{s} \text{ (meaningless)}$$

$$t = 3.28\text{s}$$

$$v(t) = 10 \frac{\text{m}}{\text{s}} - \left(9.81 \frac{\text{m}}{\text{s}^2}\right)t$$

$$v(3.28\text{s}) = 10 \frac{\text{m}}{\text{s}} - \left(9.81 \frac{\text{m}}{\text{s}^2}\right)(3.28\text{s})$$

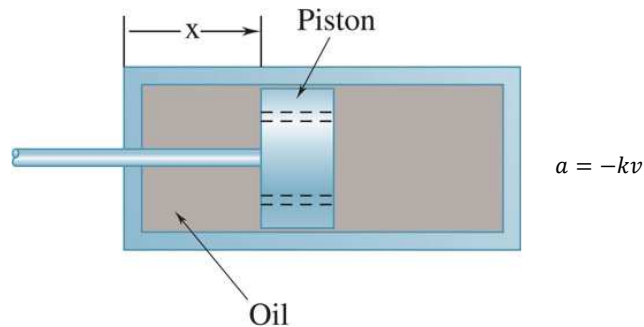
$$v = -22.2 \frac{\text{m}}{\text{s}}$$

REFLECT and THINK:

When the acceleration is constant, the velocity changes linearly, and the position is a quadratic function of time.



Sample Problem: Shock Absorption



STRATEGY:

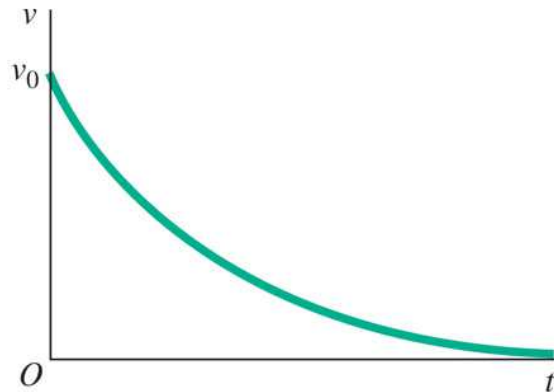
1. Integrate $a = dv/dt = -kv$ to find $v(t)$.
2. Integrate $v(t) = dx/dt$ to find $x(t)$.
3. Integrate $a = v dv/dx = -kv$ to find $v(x)$.

A mountain bike shock mechanism used to provide shock absorption consists of a piston that travels in an oil-filled cylinder. As the cylinder is given an initial velocity v_0 , the piston moves and oil is forced through orifices in piston, causing piston and cylinder to decelerate at rate proportional to their velocity.

Determine $v(t)$, $x(t)$, and $v(x)$.



Sample Problem: Shock Absorption

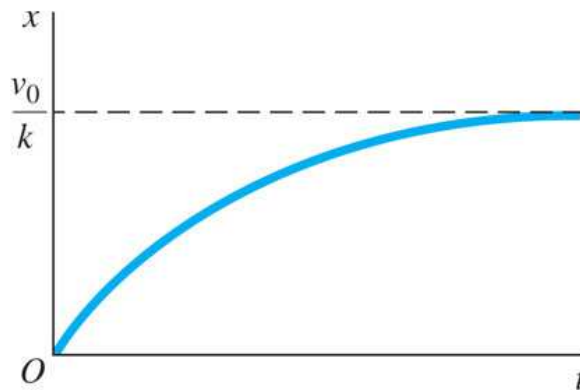


MODELING and ANALYSIS:

Integrate $a = dv/dt = -kv$ to find $v(t)$.

$$a = \frac{dv}{dt} = -kv \quad \int_{v_0}^v \frac{dv}{v} = -k \int_0^t dt \quad \ln \frac{v(t)}{v_0} = -kt$$

$$v(t) = v_0 e^{-kt}$$



Integrate $v(t) = dx/dt$ to find $x(t)$.

$$v(t) = \frac{dx}{dt} = v_0 e^{-kt}$$

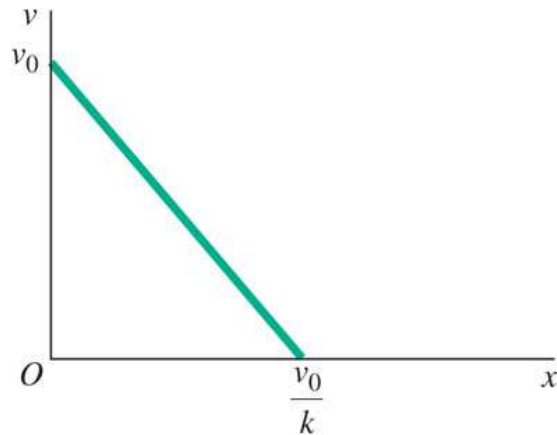
$$\int_0^x dx = v_0 \int_0^t e^{-kt} dt$$

$$x(t) = v_0 \left[-\frac{1}{k} e^{-kt} \right]_0^t$$

$$x(t) = \frac{v_0}{k} (1 - e^{-kt})$$



Sample Problem: Shock Absorption



Integrate $a = v \, dv/dx = -kv$ to find $v(x)$.

$$a = v \frac{dv}{dx} = -kv \quad dv = -kdx$$

$$\int_{v_0}^v dv = -k \int_0^x dx$$

$$v - v_0 = -kx$$

$$v = v_0 - kx$$

Alternatively,

$$\text{with } x(t) = \frac{v_0}{k} (1 - e^{-kt})$$

$$\text{and } v(t) = v_0 e^{-kt} \text{ or } e^{-kt} = \frac{v(t)}{v_0}$$

$$\text{then } x(t) = \frac{v_0}{k} \left(1 - \frac{v(t)}{v_0} \right)$$

$$v = v_0 - kx$$



Sample Problem: Shock Absorption

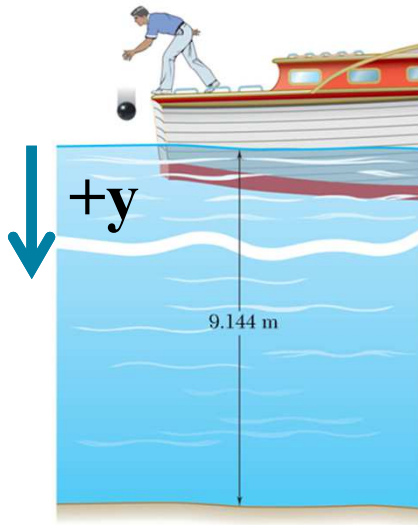
REFLECT and THINK:

You could have solved part *c* by eliminating *t* from the answers obtained for parts *a* and *b*. You could use this alternative method as a check. From part *a*, you obtain $e^{-kt} = v/v_0$; substituting into the answer of part *b*, you have:

$$x = \frac{v_0}{k} (1 - e^{-kt}) = \frac{v_0}{k} \left(1 - \frac{v}{v_0} \right) \qquad v = v_0 - kx \qquad (\text{checks})$$



Group Problem Solving: Bowling Ball



A bowling ball is dropped from a boat so that it strikes the surface of a lake with a speed of 8 m/s. Assuming the ball experiences a downward acceleration of $a = 3 - 0.1v^2$ when in the water, determine the velocity of the ball when it strikes the bottom of the lake. (a and v expressed in m/s^2 and m/s respectively)

Which integral should you choose?

(a) $\int_{v_0}^v dv = \int_0^t a(t) dt$

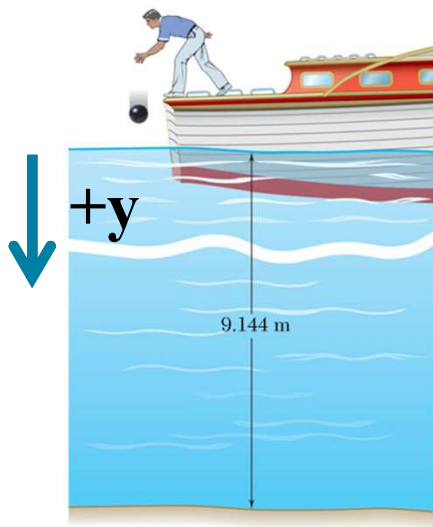
(c) $\int_{v_0}^v v dv = \int_{x_0}^x a(x) dx$

(b) $\int_{x_0}^x dx = \int_{v_0}^v \frac{v dv}{a(v)}$

(d) $\int_{v_0}^v \frac{dv}{a(v)} = \int_0^t dt$

Group Problem Solving: Bowling Ball

When will the bowling ball start slowing down?



A bowling ball is dropped from a boat so that it strikes the surface of a lake with a speed of 8 m/s. Assuming the ball experiences a downward acceleration of $a = 3 - 0.1v^2$ when in the water, determine the velocity of the ball when it strikes the bottom of the lake.

The velocity would have to be high enough for the $0.1 v^2$ term to be bigger than 3.



Group Problem Solving: Car Acceleration



The car starts from rest and accelerates according to the relationship

$$a = 3 - 0.001v^2$$

It travels around a circular track that has a radius of 200 meters. Calculate the velocity of the car after it has travelled halfway around the track. What is the car's maximum possible speed?

STRATEGY:

1. Determine the proper kinematic relationship to apply (is acceleration a function of time, velocity, or position?)
2. Determine the total distance the car travels in one-half lap
3. Integrate to determine the velocity after one-half lap



Group Problem Solving: Car Acceleration

Given: $a = 3 - 0.001v^2$
 $v_o = 0, r = 200 \text{ m}$

Find: v after $\frac{1}{2}$ lap
 Maximum speed

MODELING and ANALYSIS:

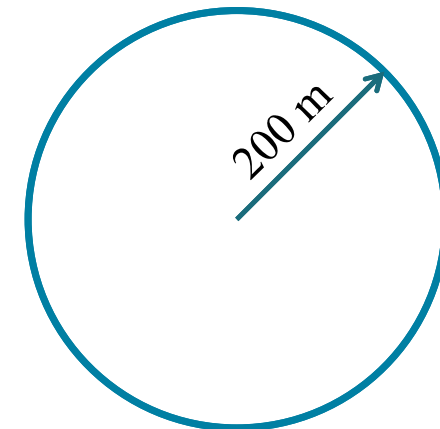
Choose the proper kinematic relationship

Acceleration is a function of velocity, and we also can determine distance. Time is not involved in the problem, so we choose:

$$v \frac{dv}{dx} = a(v) \qquad \int_{x_0}^x dx = \int_{v_0}^v \frac{v dv}{a(v)}$$

Determine total distance travelled

$$x = \pi r = 3.14(200) = 628.32 \text{ m}$$



Group Problem Solving: Car Acceleration

Determine the full integral, including limits

$$\int_{x_0}^x dx = \int_{v_0}^v \frac{v dv}{a(v)} \quad \int_0^{628.32} dx = \int_0^v \frac{v}{3 - 0.001v^2} dv$$



Evaluate the interval and solve for v

$$628.32 = -\frac{1}{0.002} \ln[3 - 0.001v^2]_0^v$$

$$628.32(-0.002) = \ln[3 - 0.001v^2] - \ln[3 - 0.001(0)]$$

$$\ln[3 - 0.001v^2] = -1.2566 + 1.0986 = -0.15802$$

Take the exponential of each side

$$3 - 0.001v^2 = e^{-0.15802}$$



Group Problem Solving: Car Acceleration

Solve for v

$$3 - 0.001v^2 = e^{-0.15802}$$



$$v^2 = \frac{3 - e^{-0.15802}}{0.001} = 2146.2$$

$$v = 46.3 \text{ m/s}$$

How do you determine the maximum speed the car can reach?

Velocity is a maximum when acceleration is zero

$$a = 3 - 0.001v^2$$

This occurs when

$$0.001v^2 = 3$$

$$v_{\max} = \sqrt{\frac{3}{0.001}}$$

$$v_{\max} = 54.8 \text{ m/s}$$



Example: Particle Motion

Given: A particle travels along a straight line to the right with a velocity of $v = (4t - 3t^2)$ m/s where t is in seconds. Also, $s = 0$ when $t = 0$.

Find: The position and acceleration of the particle when $t = 4$ s.

Plan: Establish the positive coordinate, s , in the direction the particle is traveling. Since the velocity is given as a function of time, take a derivative of it to calculate the acceleration. Conversely, integrate the velocity function to calculate the position.



Example: Particle Motion

Solution:

- 1) Take a derivative of the velocity to determine the **acceleration**.

$$a = dv/dt = d(4t - 3t^2)/dt = 4 - 6t$$

$$\Rightarrow a = -20 \text{ m/s}^2 \text{ (or in the } \leftarrow \text{ direction) when } t = 4 \text{ s}$$

- 2) **Calculate the distance** traveled in 4s by integrating the velocity using $s_0 = 0$:

$$v = ds/dt \Rightarrow ds = v dt \Rightarrow \int_{s_0}^s ds = \int_0^t (4t - 3t^2) dt$$

$$\Rightarrow s - s_0 = 2t^2 - t^3$$

$$\Rightarrow s - 0 = 2(4)^2 - (4)^3 \Rightarrow s = -32 \text{ m (or } \leftarrow \text{)}$$



Group Problem Solving: Balloon Sandbag

Given: A sandbag is dropped from a balloon ascending vertically at a constant speed of 6 m/s. The bag is released with the same upward velocity of 6 m/s at $t = 0$ s and hits the ground when $t = 8$ s.

Find: The speed of the bag as it hits the ground and the altitude of the balloon at this instant.

Plan: The sandbag is experiencing a constant downward acceleration due to gravity. Apply the formulas for constant acceleration, with $a_c = -9.81 \text{ m/s}^2$.



Group Problem Solving: Balloon Sandbag

Solution:

The bag is released when $t = 0$ s and hits the ground when $t = 8$ s.
Calculate the distance using a position equation.

$$\downarrow +s_{\text{bag}} = (s_{\text{bag}})_o + (v_{\text{bag}})_o t + (1/2) a_c t^2$$

$$s_{\text{bag}} = 0 + (-6)(8) + 0.5(9.81)(8)^2 = 265.9\text{m}$$

During $t = 8$ s, the balloon rises

$$\uparrow +s_{\text{balloon}} = (v_{\text{balloon}}) t = 6(8) = 48\text{m}$$

Therefore, altitude is of the balloon is $(s_{\text{bag}} + s_{\text{balloon}})$.

$$\text{Altitude} = 265.9 + 48 = 313.9 = 314 \text{ m.}$$



Group Problem Solving: Balloon Sandbag

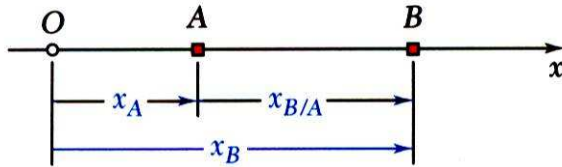
Calculate the velocity when $t = 8$ s, by applying a velocity equation.

$$\downarrow + v_{bag} = (v_{bag})_o + a_c t$$

$$v_{bag} = -6 + (9.81) 8 = 72.5 \text{ m/s } \downarrow$$



Motion of Several Particles: **Relative Motion**



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For particles moving along the same line, time should be recorded from the same starting instant and displacements should be measured from the same origin in the same direction.

$$x_{B/A} = x_B - x_A \quad \text{relative position of } B \text{ with respect to } A$$

$$x_B = x_A + x_{B/A}$$

$$v_{B/A} = v_B - v_A \quad \text{relative velocity of } B \text{ with respect to } A$$

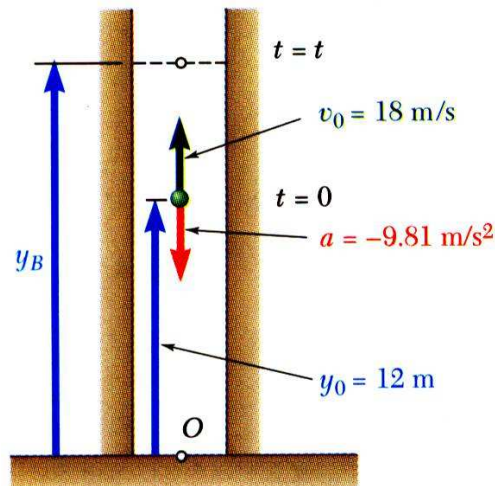
$$v_B = v_A + v_{B/A}$$

$$a_{B/A} = a_B - a_A \quad \text{relative acceleration of } B \text{ with respect to } A$$

$$a_B = a_A + a_{B/A}$$



Sample Problem: Ball and Elevator



Ball thrown vertically from 12 m level in elevator shaft with initial velocity of 18 m/s. At same instant, open-platform elevator passes 5 m level moving upward at 2 m/s.

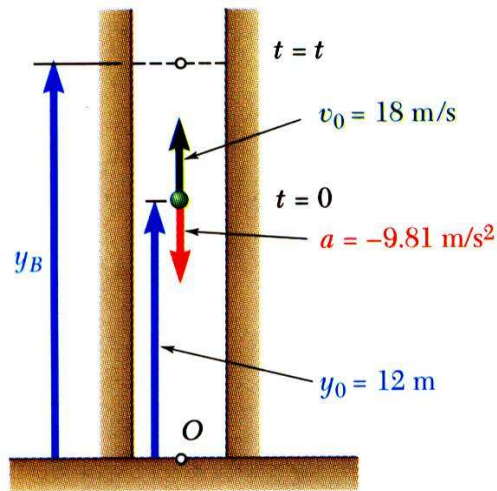
Determine (a) when and where ball hits elevator and (b) relative velocity of ball and elevator at contact.

STRATEGY:

1. Substitute initial position and velocity and constant acceleration of ball into general equations for uniformly accelerated rectilinear motion.
2. Substitute initial position and constant velocity of elevator into equation for uniform rectilinear motion.
3. Write equation for relative position of ball with respect to elevator and solve for zero relative position, i.e., impact.
4. Substitute impact time into equation for position of elevator and relative velocity of ball with respect to elevator.



Sample Problem: Ball and Elevator

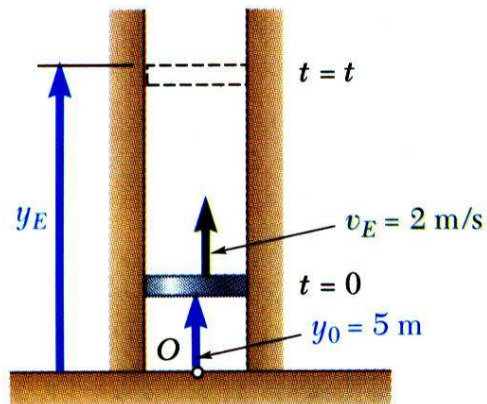


MODELING and ANALYSIS:

Substitute initial position and velocity and constant acceleration of ball into general equations for uniformly accelerated rectilinear motion.

$$v_B = v_0 + at = 18 \frac{\text{m}}{\text{s}} - \left(9.81 \frac{\text{m}}{\text{s}^2} \right) t$$

$$y_B = y_0 + v_0 t + \frac{1}{2} at^2 = 12 \text{ m} + \left(18 \frac{\text{m}}{\text{s}} \right) t - \left(4.905 \frac{\text{m}}{\text{s}^2} \right) t^2$$



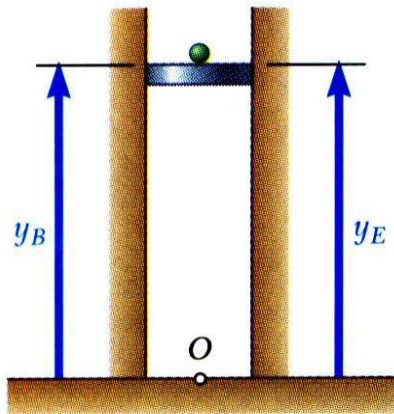
Substitute initial position and constant velocity of elevator into equation for uniform rectilinear motion.

$$v_E = 2 \frac{\text{m}}{\text{s}}$$

$$y_E = y_0 + v_E t = 5 \text{ m} + \left(2 \frac{\text{m}}{\text{s}} \right) t$$



Sample Problem: Ball and Elevator



Write equation for relative position of ball with respect to elevator and solve for zero relative position, i.e., impact.

$$y_{B/E} = (12 + 18t - 4.905t^2) - (5 + 2t) = 0$$

$$t = -0.39\text{s (meaningless)}$$

$$t = 3.65\text{s}$$

Substitute impact time into equations for position of elevator and relative velocity of ball with respect to elevator.

$$y_E = 5 + 2(3.65)$$

$$y_E = 12.3\text{m}$$

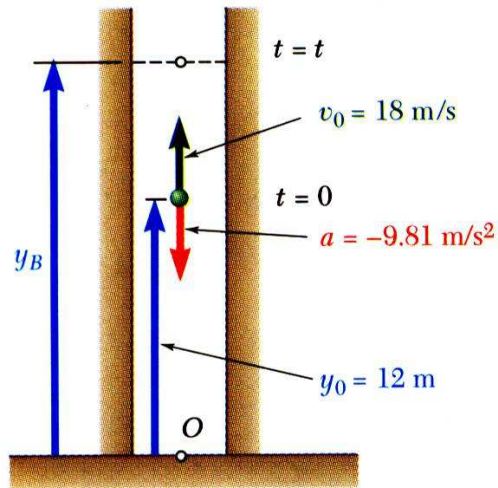
$$v_{B/E} = (18 - 9.81t) - 2$$

$$= 16 - 9.81(3.65)$$

$$v_{B/E} = -19.81 \frac{\text{m}}{\text{s}}$$



Sample Problem: Ball and Elevator



REFLECT and THINK:

The key insight is that, when two particles collide, their position coordinates must be equal. Also, although you can use the basic kinematic relationships in this problem, you may find it easier to use the equations relating a , v , x , and t when the acceleration is constant or zero.



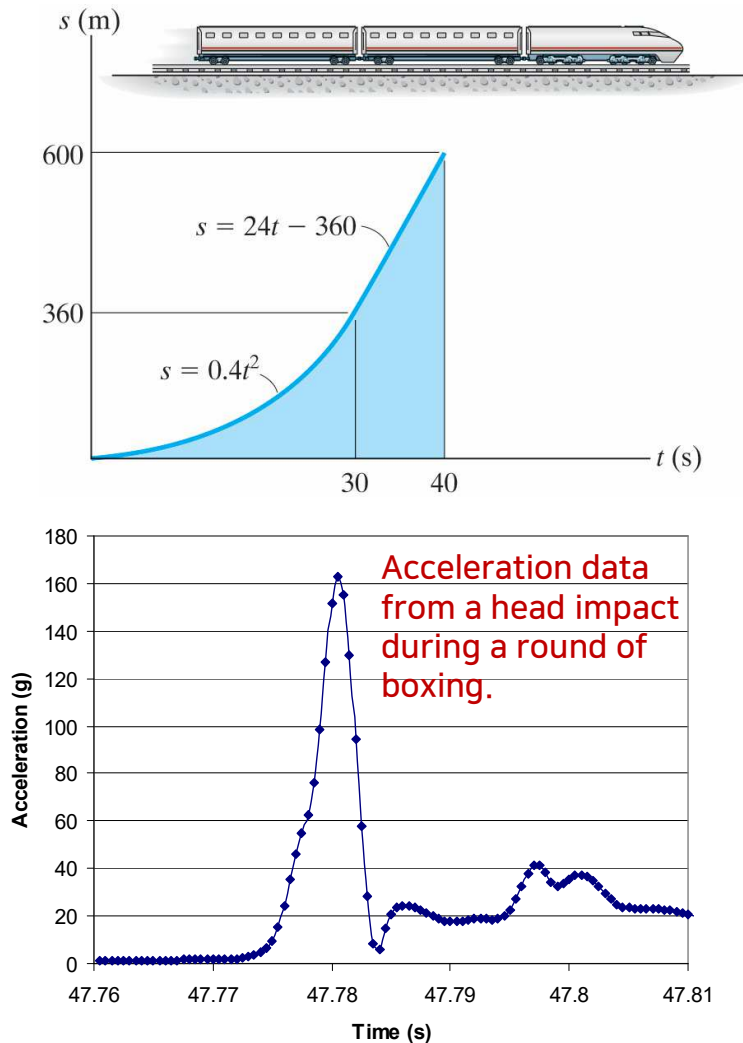
Erratic Motion: Graphical Solutions

Graphing provides a good way to handle complex motions that would be difficult to describe with formulas.

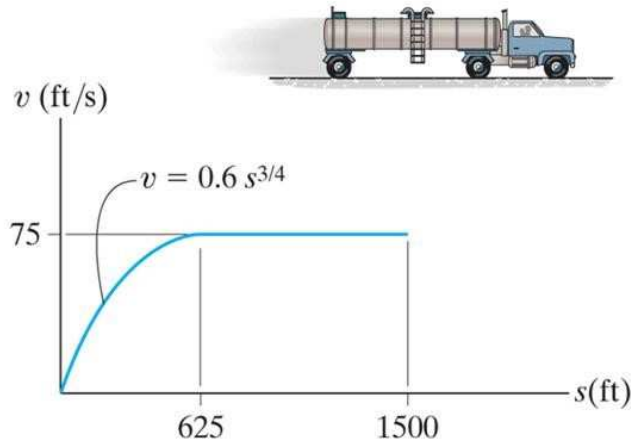
Graphs also provide a visual description of motion and reinforce the calculus concepts of differentiation and integration as used in dynamics.

Engineers often collect position, velocity, and acceleration data. Graphical solutions are often useful in analyzing these data.

The approach builds on the facts that slope and differentiation are linked and that integration can be thought of as finding the area under a curve.

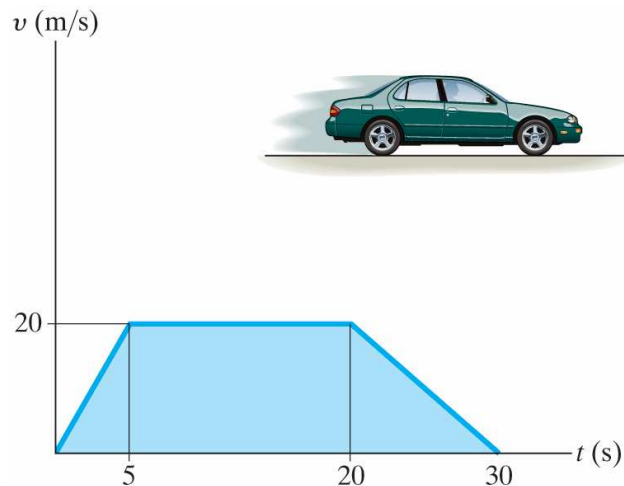


Applications



In many experiments, a velocity versus position (v-s) profile is obtained.

If we have a v-s graph for the tank truck, how can we determine its acceleration at position $s = 1500$ feet?



The velocity of a car is recorded from an experiment. The car starts from rest and travels along a straight track.

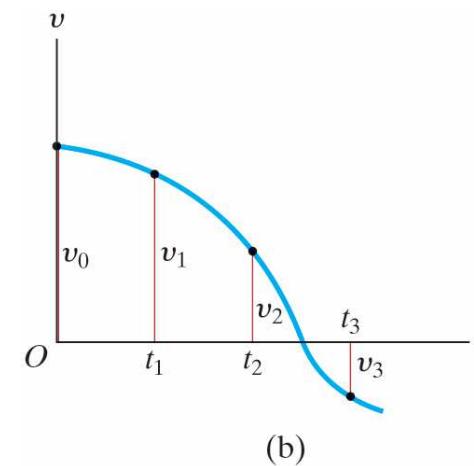
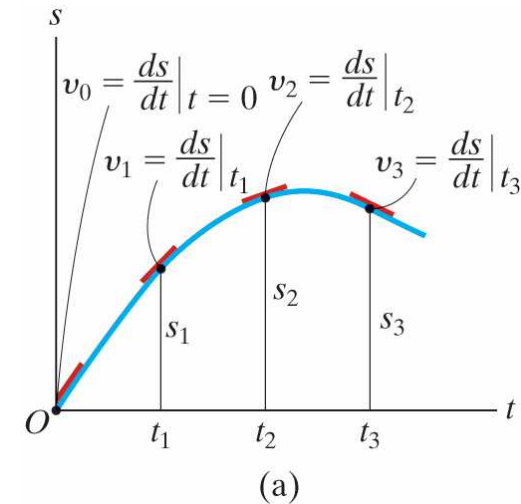
If we know the v-t plot, how can we determine the distance the car traveled during the time interval $0 < t < 30$ s or $15 < t < 25$ s?

S-T Graph

Plots of position versus time can be used to find velocity versus time curves. Finding the **slope** of the line tangent to the motion curve at any point is the **velocity** at the point

(or $v = ds/dt$).

Therefore, the v-t graph can be constructed by finding the slope at various points along the s-t graph.



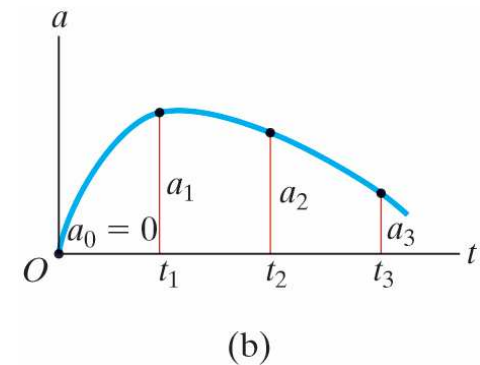
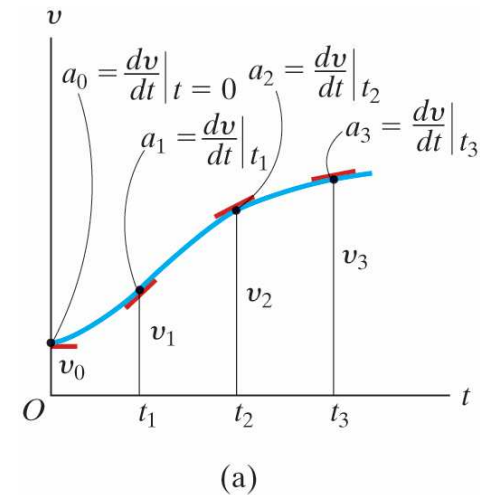
V-T Graph

Plots of velocity versus time can be used to find acceleration versus time curves. Finding the **slope** of the line tangent to the velocity curve at any point is the **acceleration** at that point

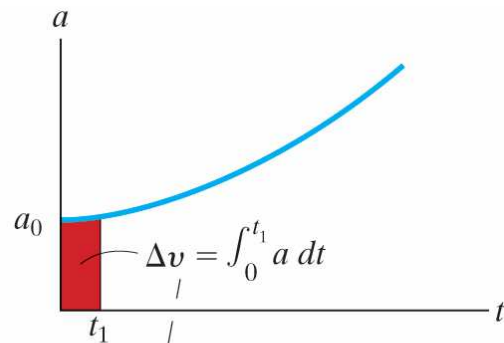
(or $a = dv/dt$).

Therefore, the v-t graph can be constructed by finding the slope at various points along the s-t graph.

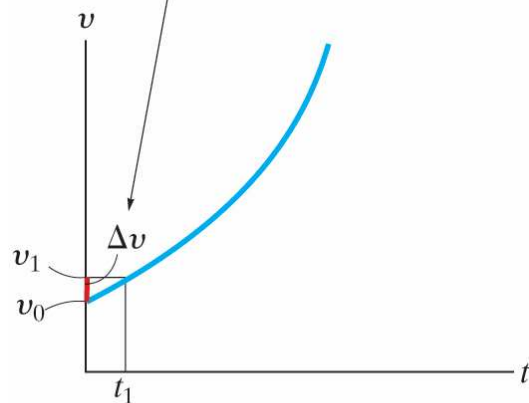
Also, the distance moved (displacement) of the particle is the area under the v-t graph during time Δt .



A-T Graph



(a)



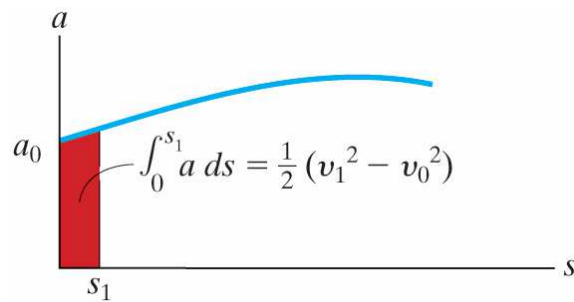
(b)

Given the acceleration versus time or a-t curve, the change in velocity (Δv) during a time period is the area under the a-t curve.

So we can construct a v-t graph from an a-t graph if we know the initial velocity of the particle.



A-S Graph

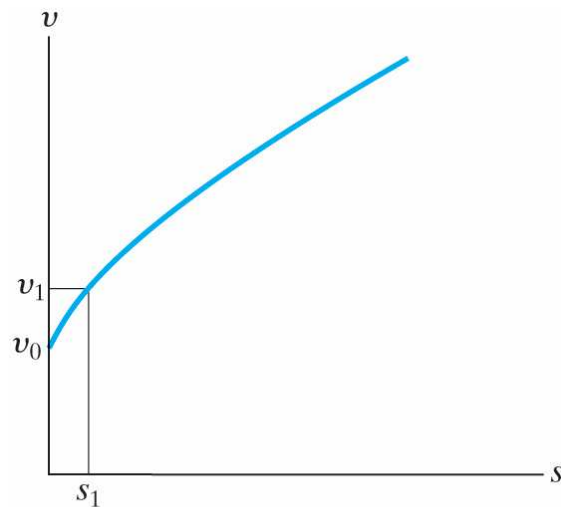


(a)

A more complex case is presented by the acceleration versus position or a-s graph. The area under the a-s curve represents **the change in velocity**

(recall $\int a \, ds = \int v \, dv$).

$$\frac{1}{2} (v_1^2 - v_0^2) = \int_{s_1}^{s_2} a \, ds = \text{area under the a-s graph}$$

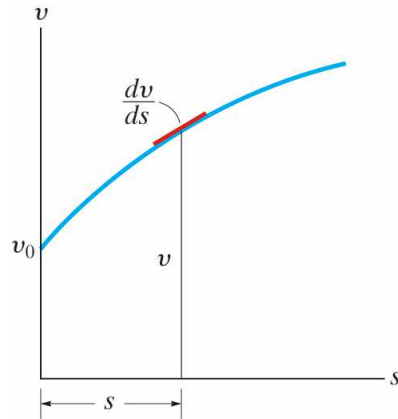


(b)

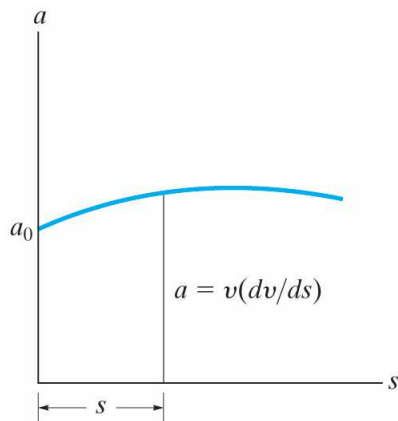
This equation can be solved for v_1 , allowing you to solve for the velocity at a point. By doing this repeatedly, you can **create a plot of velocity versus distance**.



V-S Graph



(a)



(b)

Another complex case is presented by the velocity versus distance or v-s graph. By reading the velocity v at a point on the curve and multiplying it by the slope of the curve (dv/ds) at this same point, we can obtain the acceleration at that point. Recall the formula

$$a = v \left(\frac{dv}{ds} \right).$$

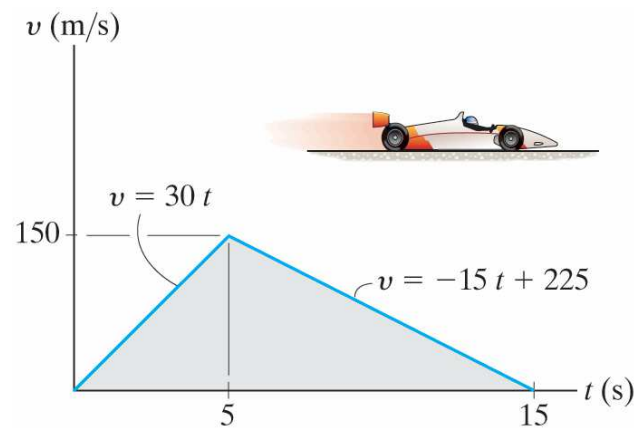
Thus, we can obtain an a-s plot from the v-s curve.



Example: Dragster

Given: The v-t graph for a dragster moving along a straight road.

Find: The a-t graph and s-t graph over the time interval shown.



What is your plan of attack for the problem?

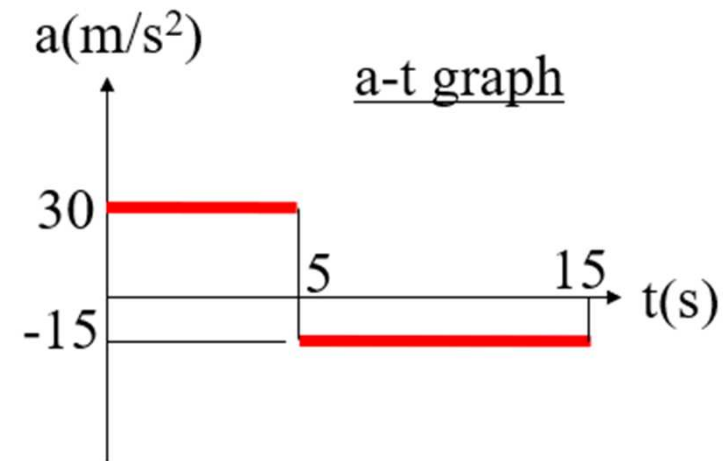
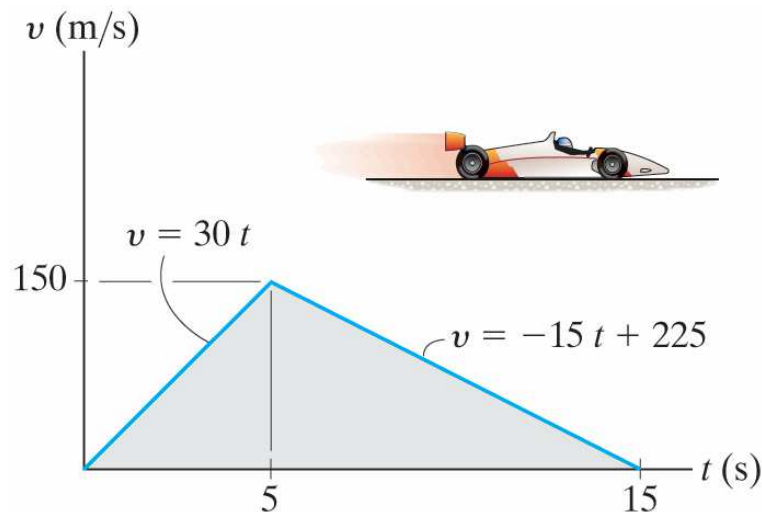


Example: Dragster

Solution: The a-t graph can be constructed by finding the slope of the v-t graph at key points. What are those?

when $0 < t < 5$ s; $v_{0-5} = ds/dt = d(30t)/dt = 30 \text{ m/s}^2$

when $5 < t < 15$ s; $v_{5-15} = ds/dt = d(-15t + 225)/dt = -15 \text{ m/s}^2$

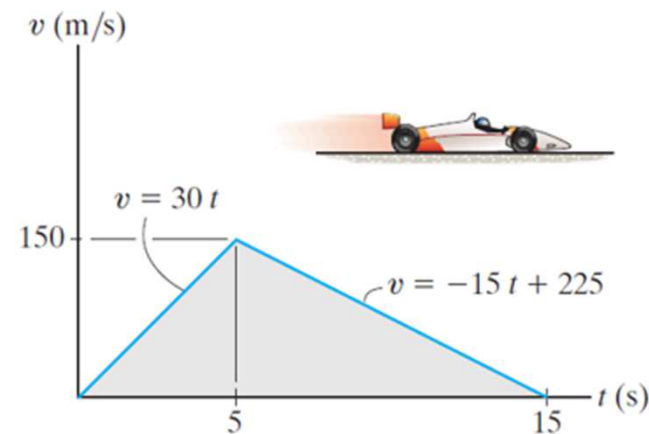
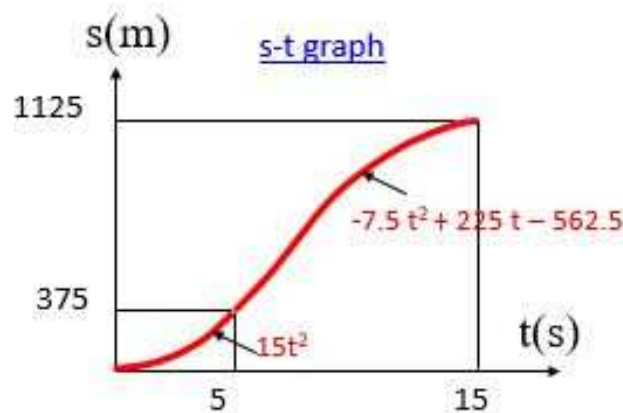


Example: Dragster

Now integrate the $v - t$ graph to build the $s - t$ graph.

when $0 < t < 5$ s; $s = \int v \, dt = [15 t^2] = 15t^2 \text{ m}$

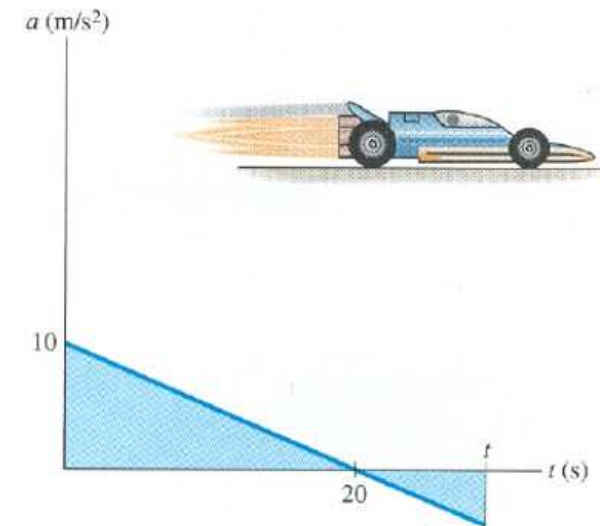
when $5 < t < 15$ s; $s - 15 (5^2) = \int v \, dt = [(-15) (1/2) t^2 + 225t]$



Concept Quiz

1. If a particle starts from rest and accelerates according to the graph shown, the particle's velocity at $t = 20$ s is

A. 200 m/s
B. 100 m/s
C. 0
D. 20 m/s



2. The particle in Problem 1 stops moving at $t =$ _____.

A. 10 s
B. 20 s
C. 30 s
D. 40 s

